

# Module 1 Forum Evidence

## Pandas preprocessing and line plots

|                       |                                               |
|-----------------------|-----------------------------------------------|
| <b>Course</b>         | CISC3015 - Introduction to Data Visualization |
| <b>Assignment</b>     | Assignment 02                                 |
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| <b>Evidence items</b> | 4                                             |

## AI use disclosure

OpenAI Codex was used to assist with code drafting, debugging, figure generation, and document presentation. The author reviewed the implementation, verified the calculations and outputs, and accepts responsibility for the submitted work.

# Lab 01 - Dataset preprocessing and exploration

The workbook was cleaned by removing empty columns, assigning descriptive field names, and computing row totals across 1980-2013. The summary confirms a complete numerical panel ready for reproducible plotting.

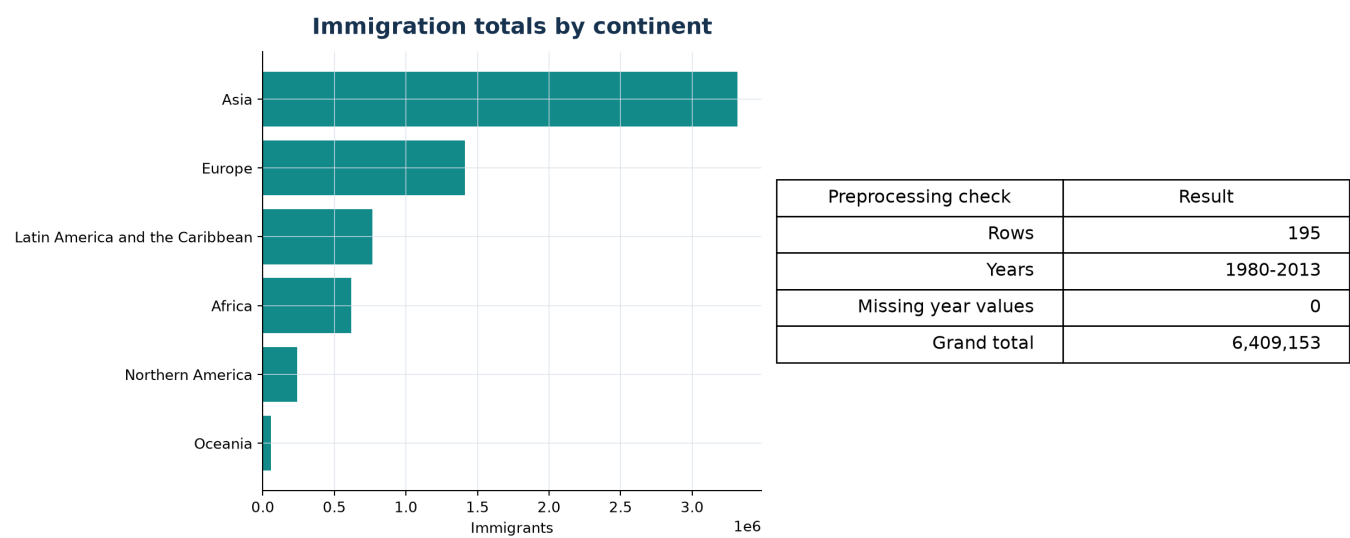


Figure 1. Dataset preprocessing and exploration. Source: analysis produced for CISC3015 Assignment 02.

## Lab 02 - Exercise 1 - Haiti immigration line plot

The line plot reveals a pronounced rise around 2010. The annotated reference line makes the social context visible without overstating causality.

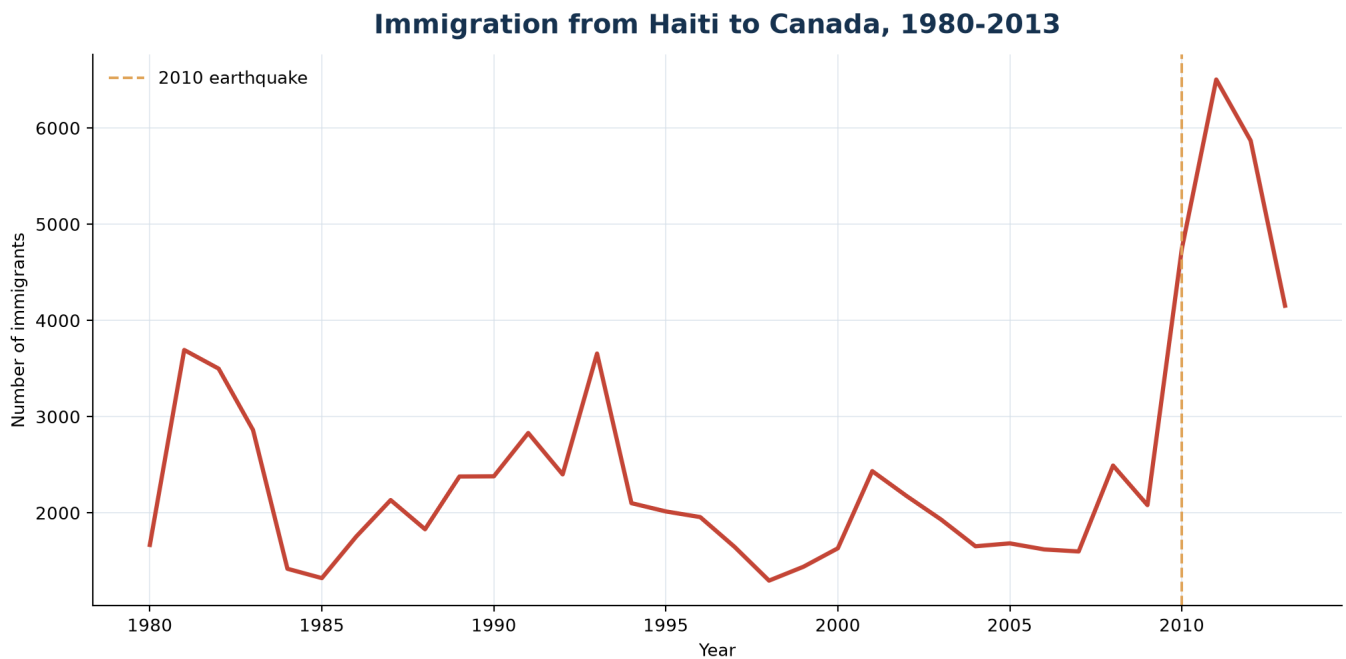


Figure 2. Haiti immigration line plot. Source: analysis produced for CISC3015 Assignment 02.

## Lab 02 - Exercise 2 - India and China comparison

Both series increase over the long run, while their year-to-year paths differ. Plotting the two countries on a shared scale supports direct comparison.

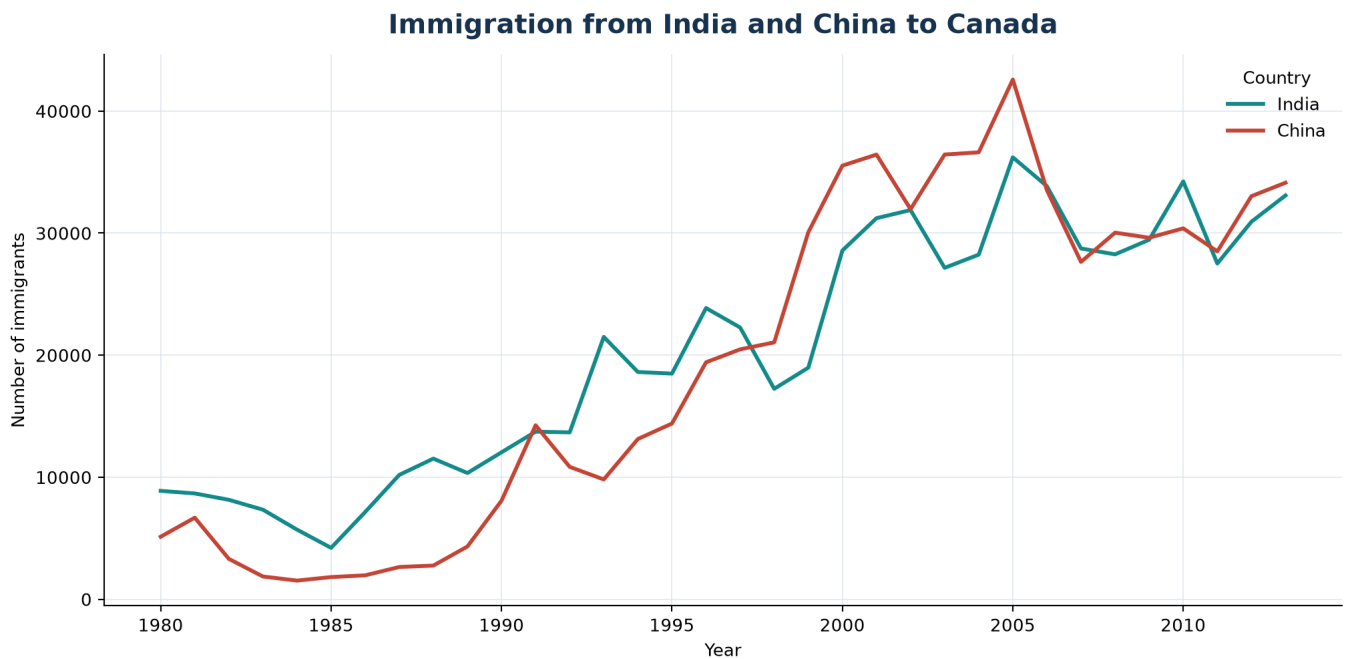


Figure 3. India and China comparison. Source: analysis produced for CISC3015 Assignment 02.

## Lab 02 - Exercise 3 - Top-five line plots

The ranking is computed from the full 1980-2013 total, then each country's annual trajectory is shown. This avoids selecting countries by a single exceptional year.

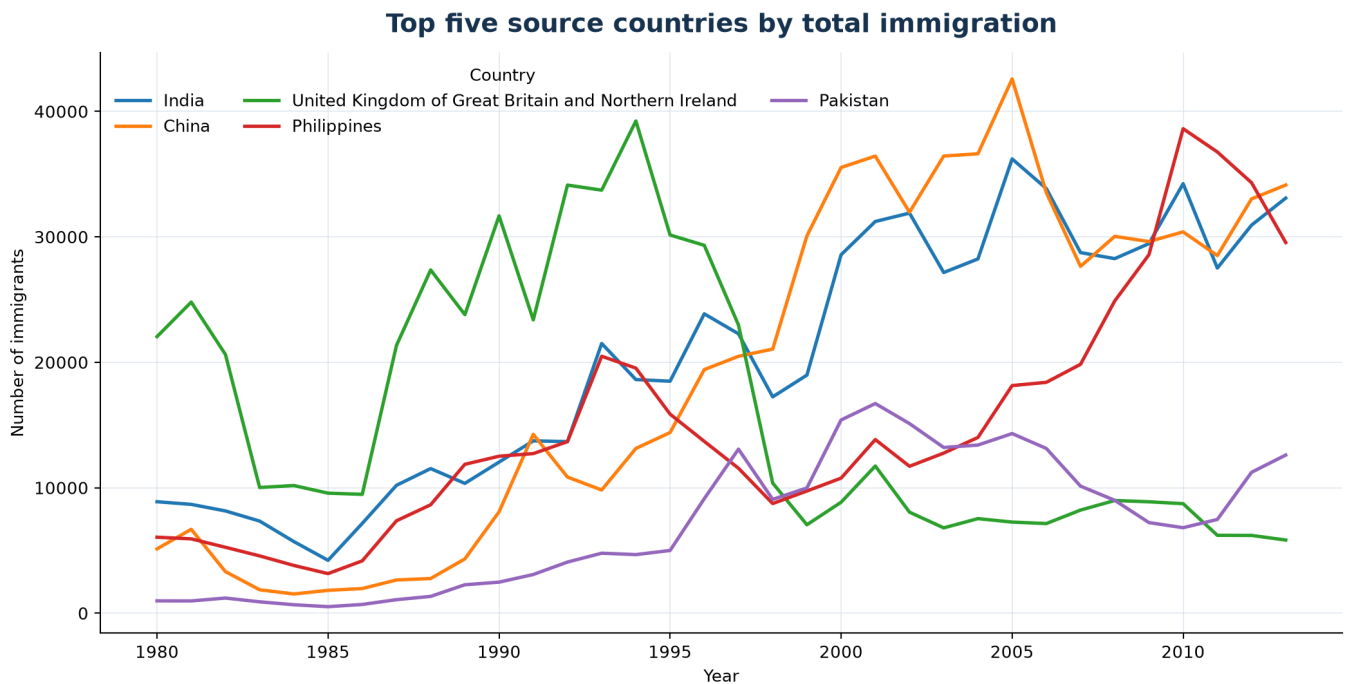


Figure 4. Top-five line plots. Source: analysis produced for CISC3015 Assignment 02.